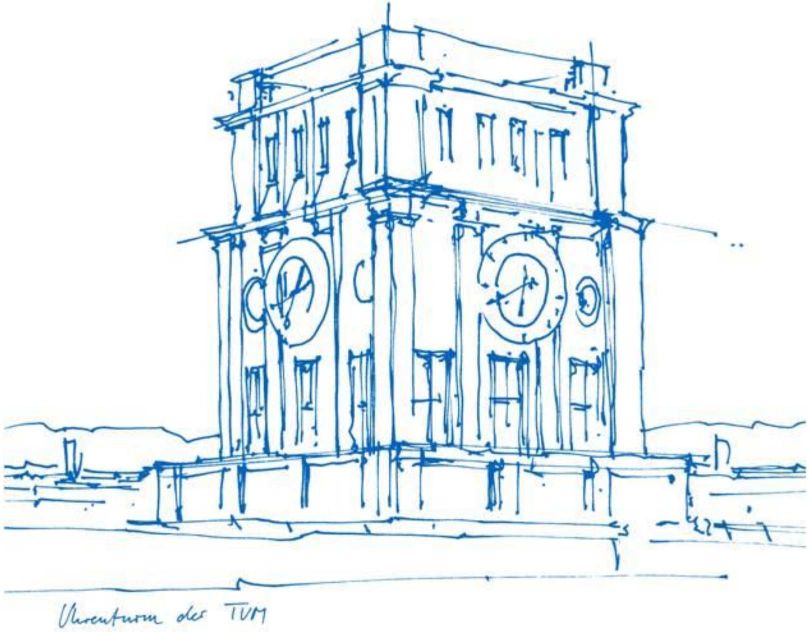
From Structural Mapping to Intention Reasoning:

A Review of LLM-based Mobile Test Migration



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Outline

- Introduction
- Background
- Methodology
- Challenges
- Future & Industry
- Conclusion

The Challenge of Heterogeneity

- ✓ Ecosystem Fragmentation: Diverse screen sizes, OS versions.
- ✓ Brittleness: Minor UI changes (e.g., renamed IDs) break traditional scripts.
- ✓ Maintenance Burden: High manual effort for test repair.

CraftDroid AppFlow

Mapping-based using semantic similarity of widgets	Core Approach	**Search-based** exploring predefined behavior graphs
Static UI attributes (XML, resource IDs, widget hierarchies)	Primary Dependencies	Supervised learning of abstract screens and labeled data
Effective **only** when UI structures are highly similar	Robustness	Robust to **layout changes** and dynamic resource IDs
Often fails on icon-based or textless UIs	UI Support	Learns abstract widgets via supervised learning
No understanding of sequential test logic; widgets are treated in isolation	Intent Reasoning	**No explicit understanding** of test intent
Poor scalability in complex applications like shopping	Main Limitations	Lack of test oracles and poor domain generalization without retraining

Why we need Large Language Models?

Paradigm shift

From rigid structural mapping → semantic intention reasoning

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The Abstraction–Planning–Grounding Pipeline

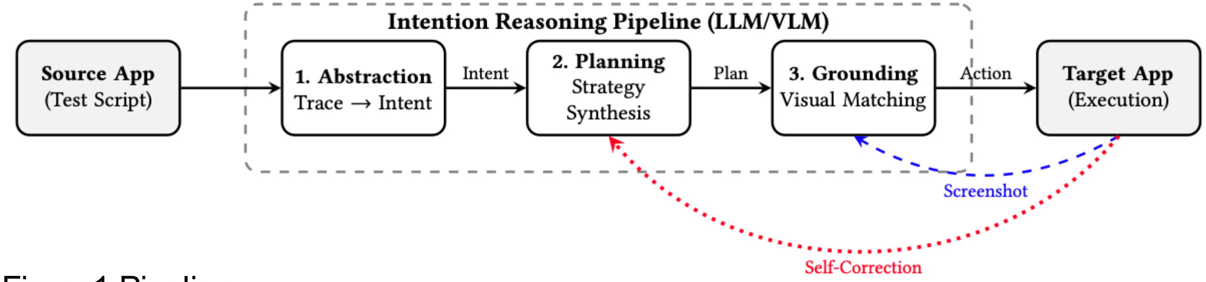
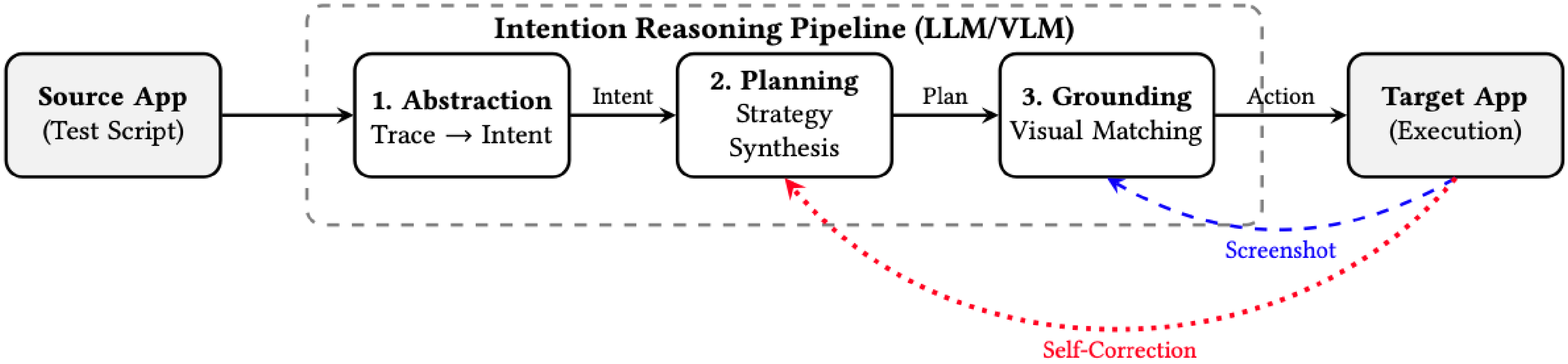


Figure1 Pipeline


Intention Reasoning Pipeline (LLM/VLM)
Source App
(Test Script)
1. Abstraction
Trace → Intent
Intent
2. Planning
Strategy
Synthesis
Plan
3. Grounding
Visual Matching
Action
Target App
(Execution)
Screenshot
Self-Correction

The Abstraction–Planning–Grounding Pipeline

- Convert low-level UI actions into platform-independent representations (Figure 2)
- Typically expressed as natural language intentions. e.g., “Verify successful login”
- Reduces dependency on UI structure and resource identifiers

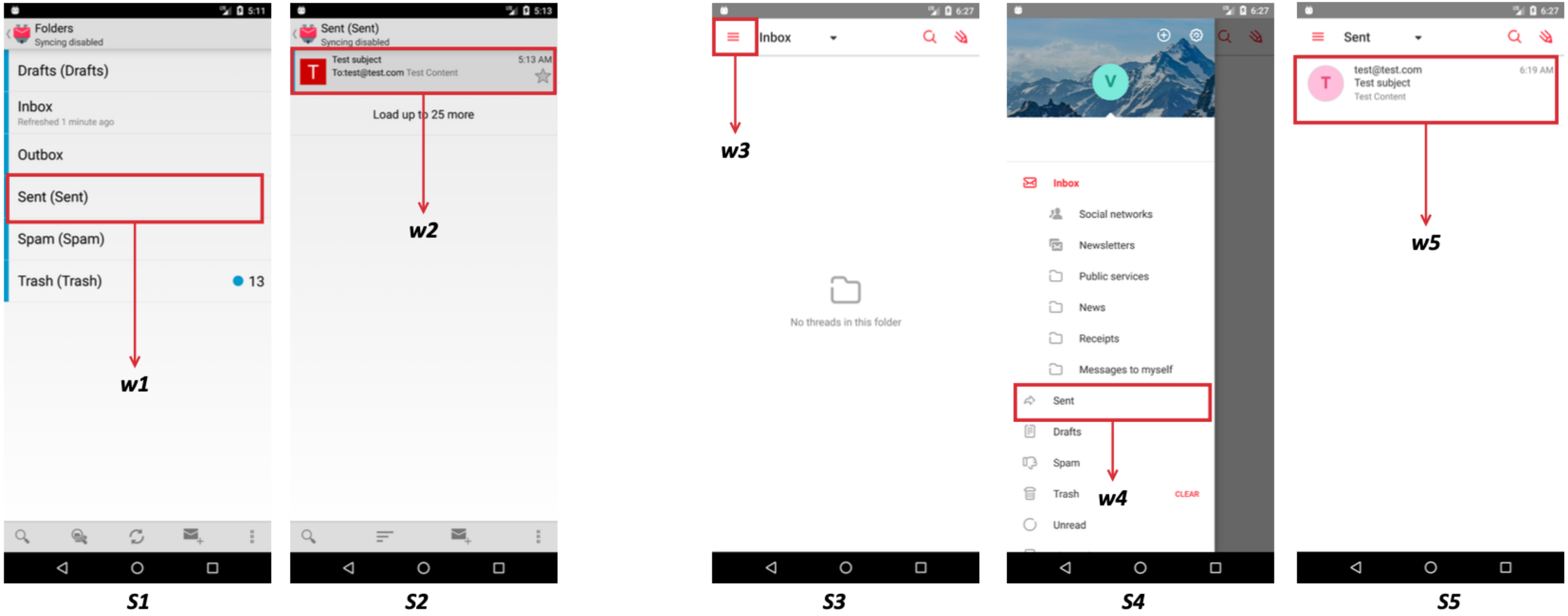


Figure2 ITEM

Source: Shaoheng Cao et al. (2025) [3]

The Abstraction–Planning–Grounding Pipeline

✓ **Key benefit:**

Enables cross-app generalization

✓ **Key risk:**

- Natural language IR introduces ambiguity
- Abstraction quality sets the upper bound of migration success

The Abstraction–Planning–Grounding Pipeline

- Translate abstract intentions into executable strategies on the target app
- Must resolve the 1-to-N mapping problem: One intent may require multiple UI actions (Figure 3)

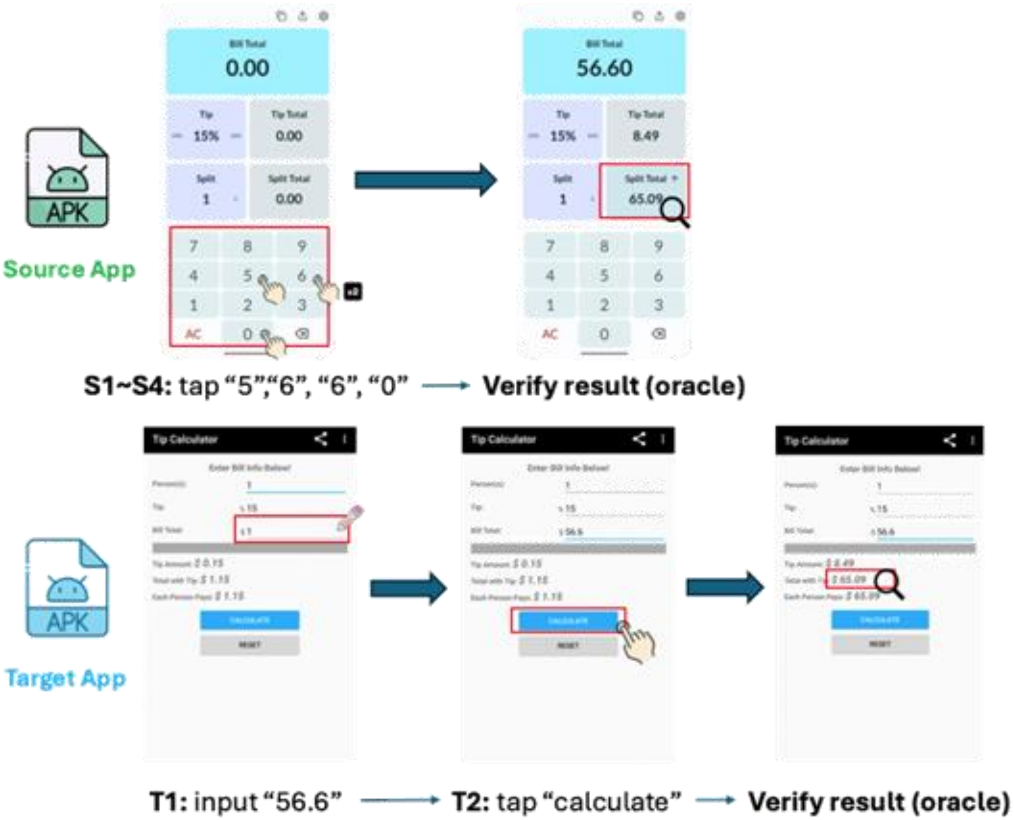


Figure3 ReuseDroid

The Abstraction–Planning–Grounding Pipeline

✓ Two dominant strategies

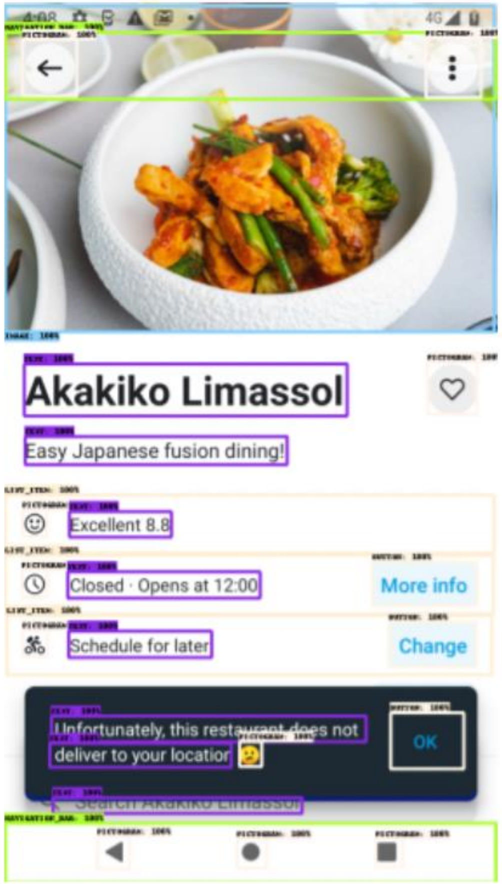
- Planner-based approaches
- Synthesis-based approaches

Planner-based Approaches | Synthesis-based Approaches

High: Preserves original semantics	Fidelity	Medium: May deviate from intent
Limited: Struggles with divergent workflows	Adaptability	**High**: Generates new execution paths
Effective **only** when UI structures are highly similar	Robustness	Robust to layout changes and dynamic resource IDs
Local execution failure	Risk	**Semantic drift**
Dead-end exploration	Typical Failure	Intent deviation

The Abstraction–Planning–Grounding Pipeline

- Map abstract actions to concrete UI elements
- Vision-Language Models (VLMs) used when metadata is unreliable (Figure 4)
- Essential for modern icon-heavy and non-standard UIs




```
IMAGE a white bowl with a chicken curry and vegetables . 0 994 4 373 (
  NAVIGATION_BAR 1 996 34 109 (
    PICTOGRAM arrow backward 36 148 43 105
    PICTOGRAM three dots 853 966 41 107)
  )
TEXT Akakiko Limassol 39 695 411 469
PICTOGRAM heart 857 959 409 467
TEXT Easy Japanese fusion dining! 40 574 493 524
LIST_ITEM 0 994 560 625 (
  PICTOGRAM happy face 35 86 577 606
  TEXT Excellent 8.8 130 339 579 607)
LIST_ITEM 1 991 628 694 (
  PICTOGRAM time 34 87 645 675
  TEXT Closed Opens at 12:00 128 518 647 676
  BUTTON More info 745 959 636 685)
LIST_ITEM 4 988 697 763 (
  PICTOGRAM 743 714 87 35
  TEXT Schedule for later 129 420 715 744
  BUTTON Change 778 957 704 754)
TEXT Unfortunately, this restaurant does not 94 733 811 839
TEXT deliver to your location 90 460 842 868
BUTTON OK 782 931 807 870
PICTOGRAM sad face 475 522 840 867
TEXT Search AkDkiKU LilliASSOT 98 603 904 921
NAVIGATION_BAR 0 997 933 999 (
  PICTOGRAM arrow backward 187 254 948 984
  PICTOGRAM a gray circle with a white background 471 532 951 983
  PICTOGRAM nav bar rect 752 809 951 982)
```

Figure4 ScreenAI

Enabling Technologies and Trade-offs

- ✓ Prompt Engineering & Context Limits
- ✓ Multimodal Context
- ✓ Self-Correction Loops

Enabling Technologies and Trade-offs

✓ **Prompt Engineering & Context Limits**

- Chain-of-Thought and in-context learning improve reasoning
- Token limits force trade-offs between context depth and complexity

✓ Multimodal Context

✓ Self-Correction Loops

Enabling Technologies and Trade-offs

✓ Prompt Engineering & Context Limits

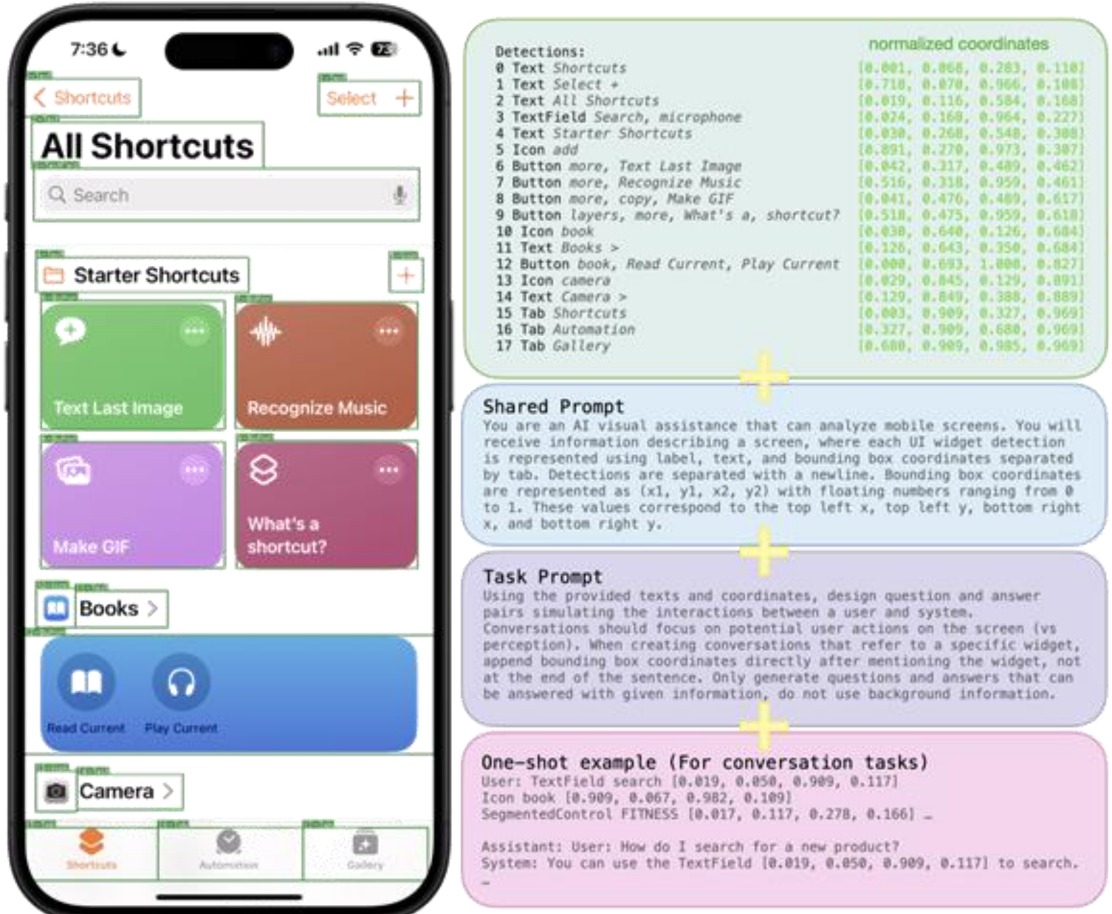
✓ **Multimodal Context** (Figure 5)

- Screenshots improve robustness
- High computational cost
- Loss of fine-grained details due to resolution constraints

✓ Self-Correction Loops

Advanced tasks overview

- Bounding box coordinates in detections are normalized
- Concatenate shared prompt and task-specific prompt
- For conversation tasks, add one-shot example
- Provide detections, prompt, and example to GPT-4
- Post-process raw output and convert to Ferret-UI format



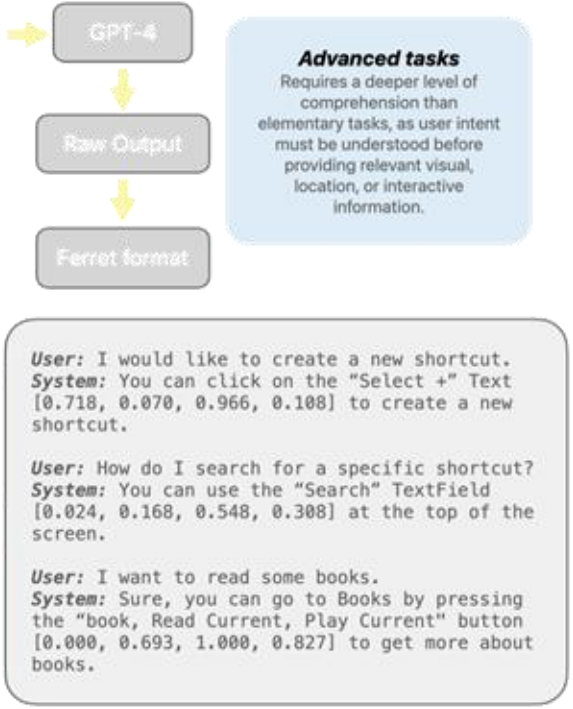


Figure5 Ferret-UI

Source: Keen You et al. (2024) [12]

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Enabling Technologies and Trade-offs

✓ Prompt Engineering & Context Limits

✓ Multimodal Context

✓ **Self-Correction Loops**

- Feedback-driven re-planning improves success rates
- Risk of infinite loops and efficiency degradation

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Open Challenges in LLM-Based Test Migration

✓ Semantic Drift in Intention Abstraction

✓ Fragility of Visual Grounding

✓ Efficiency, Stability, and Feedback Loops

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Evaluation and Reproducibility Crisis

- Most studies evaluate on small and domain-limited app sets
- LLM randomness harms reproducibility
- Existing benchmarks focus on task completion, not semantic fidelity
- Mismatch: Reaching a final UI state ≠ preserving test intent (Figure 6)

Reaching a final UI state ≠ preserving test intent

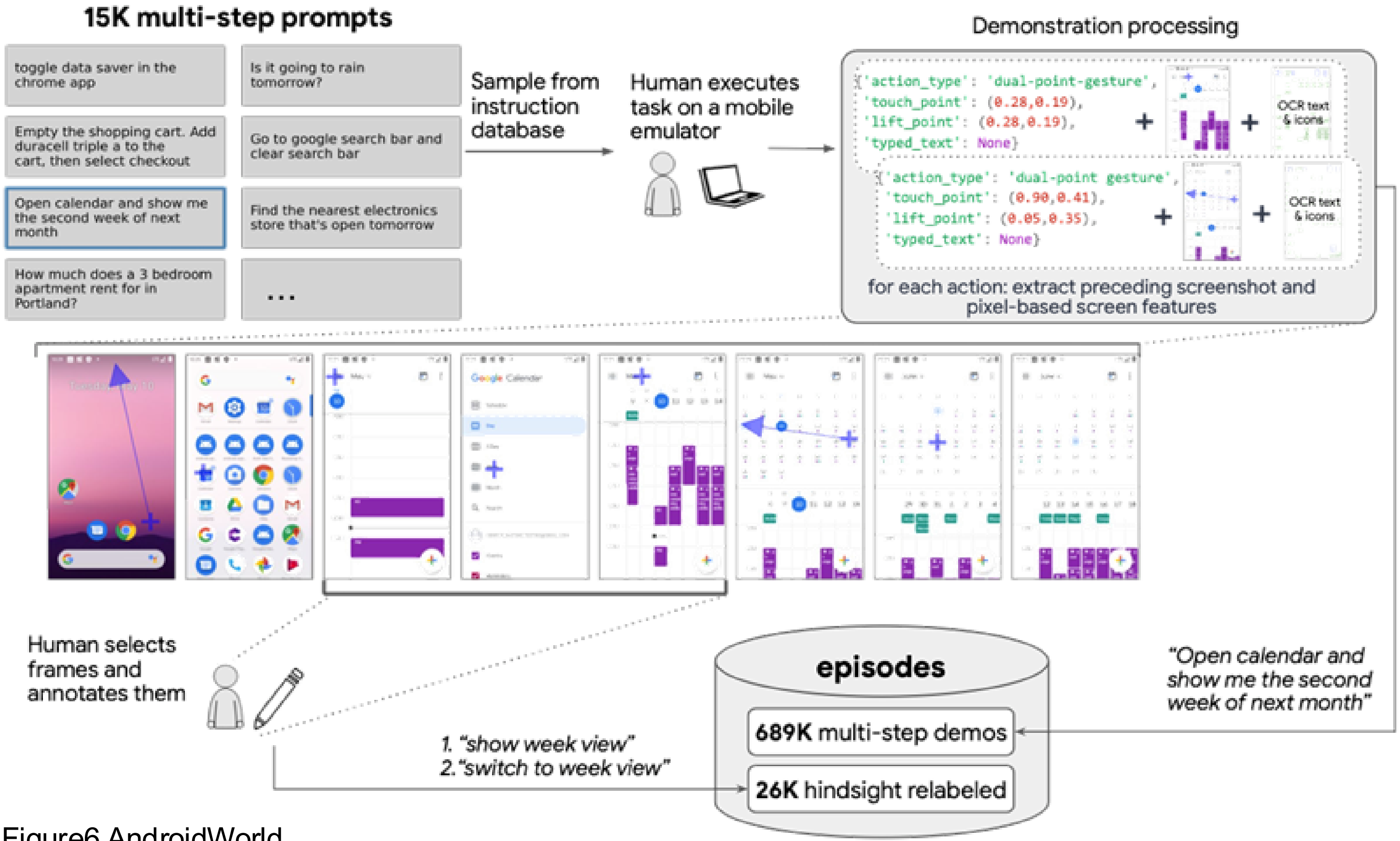

Figure6 AndroidWorld

Source: Christopher Rawles et al. (2025) [9]

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Evaluation and Reproducibility Crisis

Need:

- Step-level, intent-aware evaluation metrics

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Future Directions

✓ Long-horizon agents

✓ On-device deployment

Industrial Outlook: Amazon as a Case Study

✓ Vision-based testing scales better across heterogeneous devices

✓ Semantic test oracles replace hard-coded assertions

✓ Agent-based migration reduces maintenance costs

✓ Enables faster release cycles in large-scale systems

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Q & A

1. If VLMs can already “see” UI elements, why do we still need intention abstraction?

   Because seeing UI elements ≠ understanding test intent.

   VLMs ground actions to pixels, but intention abstraction decouples tests from UI structure, making migration robust to layout changes and enabling cross-app generalization.

2. Does LLM stochasticity undermine the reliability of automated testing?
3. Why is success rate an insufficient metric for evaluating test migration?

1. If VLMs can already “see” UI elements, why do we still need intention abstraction?
2. Does LLM randomness undermine the reliability of automated testing?

   Yes, it does, but not necessarily usefulness.

   Stochasticity introduces variance, which requires intent-aware evaluation and controlled execution, rather than eliminating LLMs altogether.
3. Why is success rate an insufficient metric for evaluating test migration?

1. If VLMs can already “see” UI elements, why do we still need intention abstraction?
2. Does LLM stochasticity undermine the reliability of automated testing?
3. Why is success rate an insufficient metric for evaluating test migration?

   Because reaching the same UI state does not guarantee preserving test intent.

   Success rate ignores semantic fidelity; migration must be evaluated at the step- and intent-level, not just final states.

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Thank you !

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